

PRAJWAL P

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on research experience in computer vision, signal processing, and multimodal machine learning. Engineered a road surface classification system at Autoliv integrating IMU, GPS, and video data using deep learning. Proficient in Python, backend development with FastAPI and Spring Boot, and full-stack application delivery.

EXPERIENCE

Research & Innovation Intern

Mar 2026 – May 2026

Autoliv, Bangalore, India

- Developed a multimodal road surface classification system integrating IMU, GPS, and video data for real-world classification.
- Processed high-frequency IMU signals (~6 kHz) using Butterworth bandpass filtering (1–80 Hz) to remove noise and sensor drift.
- Built a vision-based pipeline using CLIP to label road conditions (smooth vs. bumpy); designed hybrid labeling combining IMU variance with vision predictions.
- Extracted time-series features (RMS, rolling variance) for model input; applied overlapping windowing and augmentation to handle class imbalance.
- Architected a dual-head CNN model combining time-domain and frequency-domain features; achieved ~0.80 ROC- AUC with threshold-optimized false positive reduction.
- Visualized road quality using GPS-based heatmaps for spatial analysis of road surface data.

PROJECTS

PGRKAM AI Assistant | React.js, FastAPI, MongoDB, Groq API (LLaMA-3.3-70B), RAG

- Built a multilingual AI chatbot providing government employment guidance with domain-restricted, accurate responses.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline to ground model responses in verified domain knowledge.
- Integrated the LLaMA-3.3-70B model via Groq API, achieving low-latency inference under 1 second per query.
- Added speech-to-text and text-to-speech capabilities to support voice-based accessibility for end users.

NoteKeeper Web App | Spring Boot, React.js, MySQL

- Developed a full-stack notes management system featuring secure authentication and complete CRUD operations.
- Built RESTful APIs using MVC architecture with robust input validation and structured error handling.
- Optimized database performance through indexing strategies and efficient query design in MySQL.

EDUCATION

Presidency University, Bangalore — B.Tech in Computer Science & Engineering | CGPA: 7.10 / 10

2022 – 2026

VBR PU College, Bangalore — Class XII (PCMC)

88.5%

Maruthi Vidyalaya (ICSE), Bangalore — Class X

82.33%

SKILLS

Languages: Python, Java, JavaScript

Frameworks & Tools: React.js, Spring Boot, FastAPI, Jupyter Notebook

ML & Data: NumPy, Pandas, PyTorch/TensorFlow (basic), OpenCV

Core Concepts: Data Structures, OOP, Operating Systems, Computer Networks

Database: MySQL, MongoDB

Other: REST APIs, Git/GitHub, AWS (EC2, S3 - basics)

CERTIFICATIONS

- OCI AI Foundation Associate — Sep 2025
- AWS Fundamentals — Scaler — Jul 2025
- Advanced Python — Infosys Springboard — May 2025